

```
/* Reading the data from excel file */
```

```
PROC IMPORT DATAFILE="/home/u64288454/sasuser.v94/Telco_customer_churn.xlsx"
  OUT=churn
  DBMS=xlsx
  REPLACE;
  GETNAMES=YES;    /* Use first row as variable names */
  DATAROW=2;       /* Start reading data from row 2 (optional) */
RUN;
```

```
/* Data Preprocessing */
```

```
proc sql;
  select count(*) as rows from churn;
quit;
```

```
data churn_clean;
  set churn;

  /* Convert Total Charges (with space) from character to numeric */
  if strip('Total Charges'n) = "" then TotalCharges_num = .;
  else TotalCharges_num = input('Total Charges'n, best32.);

  /* Fix TotalCharges for new customers */
  if tenure = 0 then TotalCharges_num = 0;

  /* Create missing flag */
  Missing_TotalCharges = missing(TotalCharges_num);

  Paperless_Flag = (PaperlessBilling = "Yes");
  Partner_Flag    = (Partner = "Yes");
  Dependents_Flag = (Dependents = "Yes");

  if 'Churn Label'n = "Yes" then churn_flag = 1;
  else if 'Churn Label'n = "No" then churn_flag = 0;

  if InternetService = "No" then Internet_Type = "None";
  else Internet_Type = "Has Internet";

  if PaymentMethod = "Electronic check" then Payment_Group = "Electronic";
  else Payment_Group = "Other";

  if 'Tenure Months'n < 12 then Tenure_Group = "0-1 year";
  else if 'Tenure Months'n < 36 then Tenure_Group = "1-3 years";
  else Tenure_Group = "3+ years";
```

```
run;

proc contents data=churn_clean;
run;

proc sql;
    select count(*) as rows from churn_clean;
quit;

/* Duplicates */

proc sort data=churn_clean nodupkey;
    by customerID;
run;

proc contents data=churn_clean;
run;

/* Check for extreme values */

proc means data=churn_clean min max mean;
    var 'Monthly Charges'n TotalCharges_num 'Tenure Months'n;
run;

/* Derived Variables */

Data churn_clean;

set churn_clean;

AvgChargePerMonth = TotalCharges_num / max('Tenure Months'n,1);

High_Monthly_Charge = (MonthlyCharges > 70);

run;

/* Final sanity checks */

proc freq data=churn_clean;
    tables churn_flag Tenure_Group Internet_Type;
run;

/* Logistic Regression */

proc contents data=churn_clean;
run;

proc freq data=churn_clean;
```

```
tables 'churn value'n;
run;

proc logistic data=churn_clean;
  model 'churn value'n(event='1') =
    'Tenure Months'n
    'Monthly Charges'n
    TotalCharges_num;
run;

/* Survival Analysis */

data churn_surv;
  set churn_clean;
  event = ('churn value'n = 1); /* 1 = churn, 0 = active */
run;

proc means data=churn_surv min max;
  var 'Tenure Months'n;
run;

proc lifetest data=churn_surv plots=survival;
  time 'Tenure Months'n * event(0);
run;

proc lifetest data=churn_surv plots=survival;
  time 'Tenure Months'n * event(0);
  strata contract;
run;

data churn_surv;
  set churn_clean;

  time = 'Tenure Months'n;
  event = 'Churn Value'n;

  if time > 0 and event in (0,1);
run;

proc phreg data=churn_surv;
  model time * event(0) = 'Monthly Charges'n;
run;

proc phreg data=churn_surv;
  class contract / param=ref;
  model time * event(0) =
    'Monthly Charges'n
    contract;
run;
```

```
/* Confusion Matrix */
```

```
proc logistic data=churn_clean;  
  model 'Churn Value'n(event='1') =  
    'Tenure Months'n  
    'Monthly Charges'n  
    TotalCharges_num;  
  output out=pred p=prob_churn;  
run;
```

```
data pred;  
  set pred;  
  pred_churn = (prob_churn >= 0.5);  
run;
```

```
proc freq data=pred;  
  tables 'Churn Value'n * pred_churn / norow nocol nopercnt;  
run;
```

```
proc logistic data=churn_clean plots(only)=roc;  
  model 'Churn Value'n(event='1') =  
    'Tenure Months'n  
    'Monthly Charges'n  
    TotalCharges_num;  
run;
```